

Project Title 32:

Analyzing Emergent Behaviors in Multi-Agent LLM Systems for Educational Applications

Background & Context:

Agentic AI systems are increasingly being adopted in education to support **intelligent tutoring, personalized learning, collaborative problem-solving, and automated assessment**. In such systems, **multiple agents interact** to perform tasks such as planning lessons, solving problems, evaluating answers, and providing feedback.

As these systems grow in complexity, they may exhibit **emergent behaviors**—unexpected patterns of coordination or failure that are not explicitly programmed. These behaviors can be:

- **Positive Emergence:** Efficient collaboration, improved reasoning, adaptive learning support
- **Negative Emergence:** Infinite reasoning loops, conflicting outputs, redundant actions, or degraded performance

Understanding these emergent phenomena is crucial for designing **robust, reliable, and interpretable educational AI systems**.

Research Question:

Do multi-agent systems in educational settings exhibit emergent coordination or failure patterns, and how can these behaviors be systematically identified and categorized?

Objectives:

- To design a **multi-agent educational AI system** capable of collaborative problem-solving.
- To observe and analyze **emergent behaviors** arising from agent interactions.
- To classify these behaviors into **positive and negative emergence categories**.